

Programming Languages

- Python — ML model + backend logic
- JavaScript — frontend + smart contract interaction
- Solidity — smart contracts

ML/Data Stack

- scikit-learn (Logistic Regression, SVM, Random Forest) — mastitis risk classification, same ensemble approach as your PlantVeda work
- Pandas/NumPy — sensor data processing
- Optional: TensorFlow Lite if you want on-device inference for low-connectivity farms

Backend

- FastAPI (Python) — API layer connecting ML model, oracle, and database
- PostgreSQL/MongoDB — off-chain storage for raw sensor data (blockchain only stores hashes)

Blockchain

- Polygon (Mumbai/Amoy testnet) — cheap, fast, Ethereum-compatible, ideal for hackathons
- Solidity + Hardhat — smart contract development/testing
- Ethers.js — connecting frontend/backend to the contract
- IPFS (optional) — store larger records (vet reports, images) off-chain with hash referenced on-chain

Oracle Layer

- Custom Python script/service — collects multi-source data, applies 2-of-3 consensus check, signs, and writes to chain (this is your differentiator — build it yourself, don't just say "Chainlink")

Hardware

- Milk conductivity sensor or temperature sensor (or simulate with a manual data-entry form for hackathon demo)
- Raspberry Pi/ESP32 if you want live IoT demo — otherwise a phone camera + form works fine for judges

Frontend

- HTML/CSS/vanilla JS (your usual stack) or React if you want a slicker dashboard for farmers/buyers